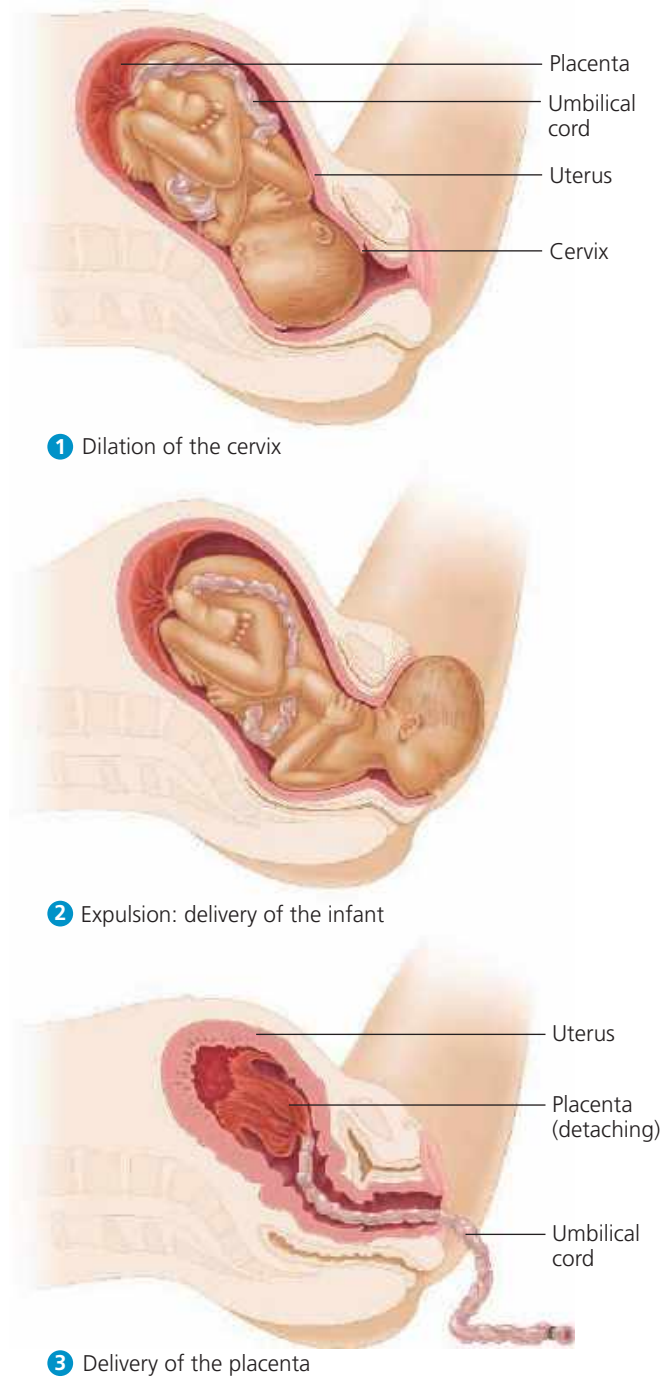


to this regulation is a positive-feedback loop (see Concept 45.2) in which uterine contractions stimulate secretion of oxytocin, which in turn stimulates further contractions.

Labor is typically described as having three stages (**Figure 46.19**). The first stage is the thinning and opening up (dilation) of the cervix. The second stage is the expulsion, or delivery, of the baby. Continuous strong contractions force the fetus out of the uterus and through the vagina. The final stage of labor is the delivery of the placenta.

▼ **Figure 46.19** The three stages of labor.



The newborn human or other mammal receives nourishment in the form of mother's milk. In response to suckling by the newborn and changes in estradiol concentration after birth, the hypothalamus signals the anterior pituitary to secrete prolactin, which stimulates the mammary glands to produce milk. Suckling also stimulates the secretion of oxytocin from the posterior pituitary, which triggers release of milk from the mammary glands (see Figure 45.15).

Maternal Immune Tolerance of the Embryo and Fetus

Pregnancy is an immunological puzzle. Because half of the embryo's genes are inherited from the father, many of the chemical markers present on the surface of the embryo are foreign to the mother. Why, then, does the mother not reject the embryo as a foreign body, as she would a tissue or organ graft from another person? One intriguing clue comes from the relationship between certain autoimmune disorders and pregnancy. For example, the symptoms of rheumatoid arthritis, an autoimmune disease of the joints, become less severe during pregnancy. Such observations suggest that the overall regulation of the immune system changes during pregnancy. Sorting out these changes and how they might protect the developing fetus is an active area of research for immunologists.

Contraception and Abortion

Contraception, the deliberate prevention of pregnancy, can be achieved in a number of ways. Some contraceptive methods prevent gamete development or release from female or male gonads; others prevent fertilization by keeping sperm and egg apart; and still others prevent implantation of an embryo. For complete information on contraceptive methods, you should consult a health-care provider. The following brief introduction to the biology of the most common methods and the corresponding diagram in **Figure 46.20** make no pretense of being a contraception manual.

Fertilization can be prevented by abstinence from sexual intercourse or by any of several kinds of barriers that keep live sperm from contacting the egg. Temporary abstinence, sometimes called *natural family planning*, depends on refraining from intercourse when conception is most likely. Because the egg can survive in the oviduct for 24–48 hours and sperm for up to 5 days, a couple practicing temporary abstinence should not engage in intercourse for a number of days before and after ovulation. Furthermore, because the timing of ovulation can vary significantly, the